

7.5 OTHER TYPES

This segment is inclusive of de novo sequencing (whole-genome sequencing without any reference genome), metagenomic sequencing, and metatranscriptomic sequencing. Metatranscriptomics is a key technique in metagenomics, which allows researchers to study the function and activity of a complete set of transcripts (RNA sequences) from environmental samples.

The growing number of metagenomic research activities is the major factor driving the growth of this market segment. Over the last few years, the focus on understanding the function and relation of gut microbiome diversity, predicting resistance to specific antibiotics, understanding host-pathogen immune interactions, quantifying gene expression changes, and tracking disease progression has increased significantly.

De novo sequencing generates an initial genomic sequence of a particular organism without a reference genome. De novo sequencing services are applied in the research of animals, plants, and microorganisms, including phylogenetic studies, analysis of species diversity, genetic markers, and other genomic research. Using de novo sequencing to obtain the genomic information of microbes provides a fresh start for exploring the genetic structure and functions, studying the evolutionary origin of microbial populations, as well as developing potential applications across medicine, disease modeling, agriculture, and the environment.

TABLE 45 OTHER MICROBIOME SEQUENCING SERVICES MARKET, BY REGION, 2021–2028 (USD MILLION)

Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
North America	12.39	13.88	15.54	17.41	19.53	21.97	24.80	28.12	12.6%
Europe	6.81	7.62	8.52	9.53	10.66	11.97	13.48	15.23	12.3%
Asia Pacific	1.91	2.17	2.46	2.79	3.16	3.59	4.08	4.65	13.6%
Latin America	0.44	0.47	0.50	0.53	0.57	0.61	0.65	0.69	6.6%
Middle East and Africa	0.31	0.32	0.34	0.35	0.37	0.39	0.40	0.42	4.6%
Total	21.85	24.46	27.36	30.62	34.29	38.52	43.41	49.11	12.4%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 46 NORTH AMERICA: OTHER MICROBIOME SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)

Country	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
US	11.23	12.59	14.11	15.82	17.76	20.00	22.60	25.65	12.7%
Canada	1.16	1.29	1.43	1.59	1.77	1.97	2.20	2.47	11.5%
Total	12.39	13.88	15.54	17.41	19.53	21.97	24.80	28.12	12.6%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 47 EUROPE: OTHER MICROBIOME SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Germany	1.46	1.65	1.86	2.10	2.37	2.68	3.04	3.47	13.3%
UK	1.31	1.48	1.68	1.90	2.14	2.43	2.77	3.17	13.6%
France	1.04	1.17	1.31	1.46	1.64	1.84	2.08	2.35	12.5%
Italy	0.29	0.33	0.37	0.41	0.45	0.51	0.57	0.64	11.9%
Spain	0.23	0.25	0.28	0.31	0.34	0.38	0.43	0.48	11.4%
Rest of Europe	2.48	2.74	3.03	3.36	3.71	4.12	4.58	5.12	11.0%
Total	6.81	7.62	8.52	9.53	10.66	11.97	13.48	15.23	12.3%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 48 ASIA PACIFIC: OTHER MICROBIOME SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
China	0.80	0.92	1.04	1.19	1.35	1.53	1.75	2.00	14.0%
Japan	0.37	0.43	0.49	0.56	0.64	0.73	0.83	0.95	14.3%
India	0.05	0.05	0.06	0.07	0.08	0.09	0.10	0.11	13.2%
Rest of Asia Pacific	0.68	0.77	0.87	0.97	1.10	1.23	1.39	1.57	12.7%
Total	1.91	2.17	2.46	2.79	3.16	3.59	4.08	4.65	13.6%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 49 LATIN AMERICA: OTHER MICROBIOME SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Brazil	0.21	0.22	0.24	0.25	0.27	0.29	0.31	0.33	6.9%
Rest of Latin America	0.23	0.25	0.27	0.28	0.30	0.32	0.34	0.36	6.4%
Total	0.44	0.47	0.50	0.53	0.57	0.61	0.65	0.69	6.6%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis